

1. 令 $\mathbf{Q}(i) = \{a+bi \mid a, b \in \mathbf{Q}\}$, 证明 $\mathbf{Q}(i)$ 是一个数域。

1. $0, 1 \in \mathbf{Q}(i)$

令 $a, b \in \mathbf{Q}(i)$, 则设 $a = x_1 + y_1 i$, $b = x_2 + y_2 i$, $a \pm b = (x_1 \pm x_2) + (y_1 \pm y_2)i \in \mathbf{Q}(i)$

$$ab = (x_1 x_2 - y_1 y_2) + (x_1 y_2 + y_1 x_2)i \in \mathbf{Q}(i)$$

设 $b \neq 0$, 即 $x_2, y_2 \neq 0$, 则 $\frac{a}{b} = \frac{(x_1 x_2 + y_1 y_2) + (x_2 y_1 - x_1 y_2)i}{x_2^2 + y_2^2} \in \mathbf{Q}(i)$ 综上, $\mathbf{Q}(i)$ 为一个数域

2. 令

$$F = \left\{ \frac{a_0 + a_1 e + \cdots + a_n e^n}{b_0 + b_1 e + \cdots + b_m e^m} \mid n, m \text{ 为任意非负整数}, a_i, b_j \in \mathbf{Z}, \right. \\ \left. 0 \leq i \leq n, 0 \leq j \leq m. \right\}$$

证明 F 是一个数域, 其中 e 是自然对数的底。

2. $a_0 = a_1 = \cdots = a_n = 0$ 时, 且 $b_0, b_1, \cdots, b_m \neq 0$ 时, 值为 0.

$m = n$, $a_0 = b_0, a_1 = b_1, \cdots, a_n = b_m$ 时, 且 $a_1, a_2, \cdots, a_n \neq 0$ 时, 值为 1.

从而 $0, 1 \in F$.

设 $a, b \in F$. 设 $a = \frac{p_n(e)}{q_m(e)}$, $b = \frac{p_m(e)}{q_m(e)}$ 则 $a \pm b = \frac{p_n(e)q_m(e) \pm q_m(e)p_m(e)}{q_m(e)q_m(e)} = \frac{x_{n+m_2}(e) \pm y_{m_1 m_2}(e)}{q_{m_1+m_2}(e)} \in F$.

$ab \in F$. 设 $b \neq 0$. 则 $\frac{a}{b} = \frac{p_n(e)q_m(e)}{q_m(e)p_m(e)} \in F$. 从而 F 为一个数域.

补充题一

1. 解下列线性方程组。

$$\begin{cases} (1+a_1)x_1 + x_2 + x_3 + \cdots + x_n = b_1 \\ x_1 + (1+a_2)x_2 + x_3 + \cdots + x_n = b_2 \\ \cdots \quad \cdots \quad \cdots \quad \cdots \quad \cdots \\ x_1 + x_2 + x_3 + \cdots + (1+a_n)x_n = b_n \end{cases}$$

其中 $a_i \neq 0, i=1, 2, \cdots, n$, 且 $\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n} \neq -1$ 。

$$1. \text{ 令 } \sum_{i=1}^n x_i = y, \text{ 则原方程化为 } \begin{cases} y + a_1 x_1 = b_1 \text{ ①} \\ y + a_2 x_2 = b_2 \text{ ②} \\ \vdots \\ y + a_n x_n = b_n \text{ ④} \end{cases} \quad \text{①} \cdot \frac{1}{a_1} + \text{②} \cdot \frac{1}{a_2} + \cdots + \text{④} \cdot \frac{1}{a_n}, \text{ 得}$$

$$\left(\frac{1}{a_1} + \cdots + \frac{1}{a_n} \right) y + y = \frac{b_1}{a_1} + \cdots + \frac{b_n}{a_n}$$

$$\text{即 } \left(\frac{1}{a_1} + \cdots + \frac{1}{a_n} + 1 \right) y = \frac{b_1}{a_1} + \cdots + \frac{b_n}{a_n}$$

$$y = \frac{\sum_{i=1}^n \frac{b_i}{a_i}}{\sum_{i=1}^n \frac{1}{a_i} + 1}$$

$$\text{即 } \left(\frac{1}{a_1} + \dots + \frac{1}{a_n} + 1 \right) y = \frac{b_1}{a_1} + \dots + \frac{b_n}{a_n} \quad y = \frac{\frac{b_1}{a_1} + \dots + \frac{b_n}{a_n}}{\sum_{i=1}^n \frac{1}{a_i} + 1}$$

$$\therefore x_m = \frac{b_m - y}{a_m} = \frac{b_m - \frac{\sum_{j=1}^n \frac{b_j}{a_j}}{\sum_{i=1}^n \frac{1}{a_i} + 1}}{a_m} \quad (1 \leq m \leq n)$$

2. 解下列线性方程组。

$$\begin{cases} x_1 + 2x_2 + 3x_3 + \dots + (n-1)x_{n-1} + nx_n = b_1 \\ nx_1 + x_2 + 2x_3 + \dots + (n-2)x_{n-1} + (n-1)x_n = b_2 \\ \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \\ 2x_1 + 3x_2 + 4x_3 + \dots + nx_{n-1} + x_n = b_n \end{cases}$$

2. 上述方程相加, 得

$$\frac{(1+n)n}{2} (x_1 + x_2 + \dots + x_n) = b_1 + \dots + b_n = S. \quad \therefore \sum_{i=1}^n x_i = \frac{2S}{n(n+1)}$$

$$\text{由 } \textcircled{1} - \textcircled{2}, \text{ 得 } (1-n)x_1 + x_2 + \dots + x_n = b_1 - b_2. \quad \text{即 } (1-n)x_1 + \frac{2S}{n(n+1)} - x_1 = b_1 - b_2$$

$$\text{从而得 } x_1 = \frac{2S}{n^2(n-1)} + \frac{b_2 - b_1}{n}$$

$$\text{通解. } x_m = \frac{2S}{n^2(n-1)} + \frac{b_{m+1} - b_m}{n} \quad (1 \leq m \leq n-1)$$

$$x_n = \frac{2S}{n^2(n-1)} + \frac{b_1 - b_n}{n}$$

3. 解下列线性方程组。

$$\begin{cases} x_1 + x_2 + \dots + x_n = 1 \\ x_2 + \dots + x_n + x_{n+1} = 2 \\ \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \\ x_{n+1} + x_{n+2} + \dots + x_{2n} = n+1 \end{cases}$$

$$3. \textcircled{2} - \textcircled{1} \quad x_{n+1} - x_1 = 1$$

$$\textcircled{3} - \textcircled{2} \quad x_{n+2} - x_2 = 1$$

$$\vdots$$

$$\textcircled{n+1} - \textcircled{n} \quad x_{2n} - x_n = 1$$

综上, 方程的解为.

$$\begin{cases} x_1 = n - x_{n+2} - \dots - x_{2n} \\ x_2 = x_{n+2} - 1 \\ \vdots \\ x_n = x_{2n} - 1 \end{cases}$$

$$\text{而 } x_{n+1} = n+1 - x_{n+2} - \dots - x_{2n} \therefore \text{从而}$$

$$x_1 = x_{n+1} - 1 = n - x_{n+2} - \dots - x_{2n}$$